

Block 11: complete runtime Solana schema v2 and bounded message signing

1. Identity and frozen scope

Owner: Codex /root, agent-1. Worktree: /private/tmp/kk715-stack11-agent1.

Branch: audit/715-stack11-agent1-20260927. Target: BitHighlander/keepkey-firmware develop.

Develop base: fc1e93746132553ad98ed60f4847c8d770732bf9. Report generation predecessor: 9c1c02e991572955e34b2ee6da407b11c8bb48d1.

Code/test qualification head: e631449c18f0f26755dc79f59f9fea1d6863e8219. Completed block10 predecessor:

476afea2ce9906caa6c67bffa900b6a961fa6c4.

Python pin: aae89d378d889b3872696469fe363cf570e5ac4c. Final containing head and PR URL are recorded in the PR body.

Main master-template SHA256: 4df8723670dbd7266ed8d8756b790725fb4f0f5886d36fe2f9b7c3aece2d64e0.

Main SOP SHA256: cd2c546d41f0a723f6ce84199a3fb03661f79ab3f49fbafca08de96a7c309e89.

The owner's latest instruction requires the ENTIRE BLOCK PR against fork develop.

This supersedes the older stacked-target instruction for this PR. Develop is an ancestor; the target diff necessarily includes accepted predecessor blocks as well as all block11 source. This is a cumulative integration PR, not a ten-line repair or a claim that earlier audits re-ran over the entire release. Predecessor evidence and outstanding release gates retain their original status. No merge is requested.

Block11 includes schema versions 1/2, eight v2 argument slots, token-amount and duration display, same-runtime-signer token metadata, exact instruction coverage, companion restrictions, bounded plain-text raw message signing, AdvancedMode and session trust, and ordinary consent. Certified roots, delegate certificates, LUT proofs and production publication are excluded from runtime 7.15 scope.

Historical PR841 head e97ecd49d03e49536a0fe690bea1319879db022b and base 58526221f47a06d44ef49c9a0642136423f6556f differ in four files despite the retirement comment saying tree-empty. Four core functions are retained from that head in the accepted predecessor; the intake JSON records exact comparisons. The full block's code is visible against develop, including its retained native tests. PR841 is historical; it is not reopened and its retirement is not evidence of an empty diff.

2. Git inventory and behavior coverage

Net target inventory command: git diff --numstat fc1e93746132553ad98ed60f4847c8d770732bf9 9c1c02e991572955e34b2ee6da407b11c8bb48d1.

Every target path is listed in section 7. Counts include code, tests, dependencies, workflows, old receipts and this report. Historical counts overlap; do not sum them.

| Behavior | Code-bearing paths in the develop-target PR | Evidence |

| --- | --- | --- |

| Schema v1/v2 bounded parser and widths | include/keepkey/firmware/solana.h; lib/firmware/solana.c; unittests/firmware/solana.cpp | Native canonical/v2, unsafe-label, trailing-byte and all-prefix truncation assertions |

| Exact program/discriminator/data/account binding | lib/firmware/solana.c; unittests/firmware/solana.cpp | Coverage/mismatch/index tests and transfer-companion wire refusal with valid control |

| Runtime-only trust, token mint and same signer | lib/firmware/fsm_msg_solana.h; lib/firmware/signed_metadata.c; deps/python-keepkey | Runtime token display control and eight invalid-attestation cases |

| Session reset and policy | lib/firmware/fsm_msg_common.h; lib/firmware/storage.c; deps/python-keepkey | Initialize/ClearSession/policy tests; newly added Solana Initialize refusal |

| Plain-text predicate and exact message consent | lib/firmware/solana.c; lib/firmware/fsm_msg_solana.h; unittests/firmware/solana.cpp; deps/python-keepkey | Printable/control-byte/key-scan native tests and message-policy wire tests |

| Malformed input cannot reach schema consent | lib/firmware/fsm_msg_solana.h; deps/python-keepkey | Three malformed forms, with/without schema; before/after wire evidence |

| Transport bounds and host consumer | include/keepkey/transport/messages-solana.options; deps/python-keepkey | raw_tx 2048, schema 256, signature 64, raw message 1024; client/protobuf consumer inspected |

| Accepted predecessor stack | Remaining cumulative paths in section 7 | Existing accepted reports included; not attributed to S11-01 |

Original unit inventories:

Commit a1b63ac5e75c6792f15f8d439e452eca2f2ecb45:

13 7 include/keepkey/firmware/solana.h

93 12 lib/firmware/fsm_msg_solana.h

32 7 lib/firmware/solana.c

36 0 unittests/firmware/solana.cpp

Commit a7daa79ded5c38afef37c23d02aea6fa671f28a9:

6 0 include/keepkey/firmware/solana.h

32 25 lib/firmware/fsm_msg_solana.h

13 0 lib/firmware/solana.c

58 0 unittests/firmware/solana.cpp

Dependency pins at qualification:

a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec code-signing-keys (heads/master)

8a392f70a5d5575e3dfb35f115d4a4b27f497c deps/crypto/trezor-firmware (8a392f70a)

5fec9e6906a340be5eb3d795ec746769065b2db8 deps/device-protocol (dice-entropy-v1-83-g5fec9e6)

7888184f28509dba839e3683409443e0b5bb8948 deps/googletest (release-1.8.0-755-g7888184f)

aae89d378d889b3872696469fe363cf570e5ac4c deps/python-keepkey (v6.0.3-958-gaae89d3)

6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8 deps/qrenc/QR-Code-generator (v1.4.0-11-g6dfbfda)

71d356a1141624994cf613bd2d2583892e8e6d5a deps/sca-hardening/SecAESSTM32 (heads/keepkey)

3. Current-source reconciliation and findings

S11-01 FIXED: SolanaSignTx formerly evaluated a valid runtime schema after solana_inspectTx returned MALFORMED with partially populated instructions. That branch preceded the final malformed rejection and skipped signer validation.

The new early error zeroes the derived node and returns SyntaxError before any schema evaluation. Baseline emulator tests reached confirmation on all three schema-bearing cases; rejecting the prompt produced ActionCancelled. The fixed handler returns the malformed error with no confirmation for all six cases.

The baseline test did not approve a signature; signing reachability is source tracing, not a claim of a captured baseline signature or network acceptance.

Security closure: the sensitive operation is Ed25519 signing with the device's derived private key; host input is raw_tx plus a runtime-signed schema. The host observes protocol response and confirmation requests. Full firmware is affected; BTC does not dispatch Solana. During parsing/schema selection malformed payloads must yield no signing consent or signature. Positive controls are existing valid runtime-schema signing and token metadata reviews. Negative controls are legacy trailing bytes, missing v0 lookup count, and truncated lookup entry with/without schema. Baseline failures falsify the old guard order; fixed results prove early rejection. Artifacts are in the adjacent evidence archive and local receipt.

Parser exhaustion, safe labels, typed argument bounds and exact byte coverage were reviewed in current source. Runtime signer ID checks precede narrowing; signature verification requires exactly 64 bytes and AdvancedMode. Token display requires a matching mint, bounded decimals, valid symbol/signature and the same signer slot as the schema; otherwise it displays base units and mint. Companions cannot silently transfer value; priority fee disclosure remains in the schema review. Raw printable text rejects the actual signer key at every byte offset and retains message-byte consent. Other message formats retain AdvancedMode.

CI-TARGET-01 PENDING: develop has no KK_RELEASE_MISSING_CAPABILITIES declaration.

The inherited report validator uses the PR base as independent waiver authority, so the develop-target report gate cannot validate this staged release's ledger. The repository has no KK_ACCEPTED_WAIVER_SHA variable for manual CI either. No waiver, gate or repository setting was changed to hide this incompatibility. This is a target/qualification blocker, not a newly found signing defect. It must be resolved before calling hosted qualification green or requesting review.

4. Verification and artifact identity

- Full integrated firmware-unit: 669 tests pass on e631449c18f0f26755dc79f59ffea1d6863e8219.
- Full emulator and firmware-unit builds pass with image 7438e53933d4.
- Solana/runtime/session wire suites: 39 pass, 14 subtests pass, three skip.
- Earlier S11-01 baseline: three schema-bearing negative cases fail; schema-free controls pass.
- Earlier fixed baseline: 44 native Solana tests and 34 wire tests pass.
- Block09/10 changes are merged, preserving Python and firmware predecessor content.
- Preflight status at report generation: Pending final frozen-head preflight; do not request review yet..

- Hosted CI: pending publication on the exact report-containing head; CI-TARGET-01 remains open.
- ARM/BTC: predecessor evidence exists; fresh integrated candidate qualification pending.
- Physical-device/OLED and power-cycle verification remain release gates.

Exact skip reasons:

- test_relay_certified_v0_no_lookup_proof_reaches_signer_check: Firmware version 7.16.0 or higher is required to run this test
- test_advanced_mode_is_off_after_power_cycle: power cycle needs an emulator process this harness owns; none is bound to udp/12144 (CI runs it as a separate container). Run locally, or record the unplug/replug as manual evidence.
- test_signer_dropped_by_power_cycle: power cycle needs an emulator process this harness owns; none is bound to udp/12144 (CI runs it as a separate container). Run locally, or record the unplug/replug as manual evidence.

5. Report render and local preflight

The PDF is generated from this Markdown source. Frozen final Git inventory and PDF legibility are checked before publication; the final receipt records those checks and the containing head. The entire target diff is code-bearing; all block11 behaviors map to visible implementation/test paths above. The Python regression is published as a companion branch and pinned in this candidate.

6. Copilot checkpoint and completion

Owner authorization: exactly one run. Requests used: 0. External review pending. No block11 completion or release acceptance claimed. Do not spend the run on a repair-only or receipt-only PR. CI-TARGET-01 must be reconciled before the request. All predecessor decisions and review caveats retain their original status.

7. Complete target diff inventory

Added	Deleted	Path
---	---	---
1	2	.gitattributes
755	587	.github/workflows/ci.yml
8	8	.github/workflows/mirror-base-image.yml
275	69	.github/workflows/release.yml
6	0	.gitignore
21	0	.gitleaks.toml
5	0	.gitleaksignore
3	3	.gitmodules
58	4	CMakeLists.txt
1	0	cmake/caches/device.cmake
40	0	cmake/toolchains/mingw-w64-x86_64.cmake
19	1	deps/crypto/CMakeLists.txt
1	1	deps/crypto/trezor-firmware
1	1	deps/device-protocol
1	1	deps/python-keepkey
192	0	docs/DiceEntropy.md
154	0	docs/dice-vs-coldcard.md
94	0	docs/firmware/reviews/7.15.0-review-round2-remediation.md
124	0	docs/release/7.14.3-COMBINED-CANDIDATE.md
230	0	docs/release/REHEARSAL-SOP.md
148	0	docs/release/RELEASE-PROGRAM.md
63	0	docs/release/audit-units/715-04-review-20260923.md
-	-	docs/release/audit-units/715-04-review-20260923.pdf
61	0	docs/release/audit-units/715-05-review-20260924.md
-	-	docs/release/audit-units/715-05-review-20260924.pdf
678	0	docs/release/audit-units/715-05-skips-20260924.tsv
20	0	docs/release/audit-units/715-06-evidence-sha256-20260924.txt
82	0	docs/release/audit-units/715-06-history-20260924.tsv
-	-	docs/release/audit-units/715-06-intake-evidence-20260924.tgz
140	0	docs/release/audit-units/715-06-reconciliation-20260924.md
185	0	docs/release/audit-units/715-06-review-20260924.md
-	-	docs/release/audit-units/715-06-review-20260924.pdf

663	0	docs/release/audit-units/715-06-skips-20260924.tsv
-	-	docs/release/audit-units/715-06-validation-20260924.tgz
1086	0	docs/release/audit-units/715-06-validation-sha256-20260924.txt
144	0	docs/release/audit-units/715-07-first-copilot-retrospective-20260924.md
-	-	docs/release/audit-units/715-07-first-copilot-retrospective-20260924.pdf
81	0	docs/release/audit-units/715-07-full-audit-remediation-20260924.md
-	-	docs/release/audit-units/715-07-full-audit-remediation-20260924.pdf
112	0	docs/release/audit-units/715-07-history-20260924.tsv
62	0	docs/release/audit-units/715-07-intake-20260924.md
141	0	docs/release/audit-units/715-07-review-20260924.md
-	-	docs/release/audit-units/715-07-review-20260924.pdf
577	0	docs/release/audit-units/715-07-skips-20260924.tsv
-	-	docs/release/audit-units/715-07-validation-20260924.tgz
112	0	docs/release/audit-units/715-07-validation-sha256-20260924.txt
42	0	docs/release/audit-units/715-07b-d-resplit-mapping-20260927.md
35	0	docs/release/audit-units/715-07b-resplit-checks-20260927.json
74	0	docs/release/audit-units/715-07b-resplit-receipt-20260927.md
128	0	docs/release/audit-units/715-07b1-formatters-audit-20260926.md
-	-	docs/release/audit-units/715-07b1-formatters-audit-20260926.pdf
119	0	docs/release/audit-units/715-07b2a-groups-audit-20260927.md
-	-	docs/release/audit-units/715-07b2a-groups-audit-20260927.pdf
121	0	docs/release/audit-units/715-07b2b-iteration-audit-20260927.md
-	-	docs/release/audit-units/715-07b2b-iteration-audit-20260927.pdf
136	0	docs/release/audit-units/715-07b3r-embedded-audit-20260927.md
-	-	docs/release/audit-units/715-07b3r-embedded-audit-20260927.pdf
203	0	docs/release/audit-units/715-08-10-integration-20260927.md
-	-	docs/release/audit-units/715-08-10-integration-20260927.pdf
-	-	docs/release/audit-units/715-08-10-local-evidence-20260927.tgz
110	0	docs/release/audit-units/715-08-audit-20260927.md
-	-	docs/release/audit-units/715-08-audit-20260927.pdf
86	0	docs/release/audit-units/715-08-consolidated-audit-20260927.md
-	-	docs/release/audit-units/715-08-consolidated-audit-20260927.pdf
-	-	docs/release/audit-units/715-08-local-evidence-20260927.tgz
175	0	docs/release/audit-units/715-08-reconciliation-20260927.json
80	0	docs/release/audit-units/715-08B-audit-20260927.md
-	-	docs/release/audit-units/715-08B-audit-20260927.pdf
114	0	docs/release/audit-units/715-09-audit-20260927.md
-	-	docs/release/audit-units/715-09-audit-20260927.pdf
-	-	docs/release/audit-units/715-09-local-evidence-20260927.tgz
145	0	docs/release/audit-units/715-09-reconciliation-20260927.json
163	0	docs/release/audit-units/715-10-audit-20260927.md
-	-	docs/release/audit-units/715-10-audit-20260927.pdf
31	0	docs/release/audit-units/715-10-completion-plan-20260927.md
-	-	docs/release/audit-units/715-10-final-readiness-evidence-20260927.tgz
73	0	docs/release/audit-units/715-10-intake-20260927.md
48	0	docs/release/audit-units/715-10-reconciliation-20260927.json
523	0	docs/release/audit-units/715-11-complete-block-audit-20260927.md
-	-	docs/release/audit-units/715-11-complete-block-audit-20260927.pdf
200	0	docs/release/audit-units/715-11-intake-20260927.json
36	0	docs/release/audit-units/715-11-intake-20260927.md
-	-	docs/release/audit-units/715-11-local-evidence-20260927.tgz
77	0	docs/release/audit-units/715-11-malformed-schema.md
26	0	docs/release/audit-units/P01-002-artifact-routing.md
27	0	docs/release/audit-units/P01-003-firmware-packaging.md
12	0	docs/release/audit-units/P01-004-report-inputs.md
16	0	docs/release/audit-units/P01-008-compose-variant.md
27	0	docs/release/audit-units/P02-001-tiny-lifetime.md
14	0	docs/release/audit-units/P02-003-packet-lifetime.md
18	0	docs/release/audit-units/P06-002-ripple-display-response.md
15	0	docs/release/audit-units/P06-003-ripple-display-test.md
21	0	docs/release/audit-units/P06-005-ripple-unsupported-memo.md
12	0	docs/release/audit-units/P06-006-ripple-memo-rejection-test.md
23	0	docs/release/audit-units/decode-boundaries.md
15	0	docs/release/audit-units/eos-authorization.md
87	0	docs/release/audit-units/full-diff-audit-20260912.md

14	0	docs/release/audit-units/passphrase-transition.md
32	0	docs/release/audit-units/scope-repair.md
14	0	docs/release/audit-units/source-storage-policy.md
21	0	docs/release/audit-units/storage-capacities.md
11	0	docs/release/audit-units/storage-cipher-cleanup.md
11	0	docs/release/audit-units/storage-revocation.md
6	0	docs/release/audit-units/token-build-dependencies.md
170	0	docs/security/7.14.3-bitcoin-only-dice-audit-sop.md
235	0	docs/security/HANDOFF-ERC7730-715-FORMATTERS.md
295	0	docs/security/HANDOFF-ERC7730-ALPHA.md
209	0	docs/security/HANDOFF-ERC7730-PHASE-E.md
-	-	docs/security/evidence/rom-printf-integer-percent/01-thorchain-withdraw-25.05pct.png
-	-	docs/security/evidence/rom-printf-integer-percent/02-thorchain-sending-eth.png
17	0	docs/security/evidence/rom-printf-integer-percent/README.md
28	0	include/keepkey/board/bsd_compat.h
67	1	include/keepkey/board/confirm_sm.h
35	0	include/keepkey/board/draw.h
16	0	include/keepkey/board/keepkey_display.h
21	0	include/keepkey/board/layout.h
57	24	include/keepkey/board/messages.h
5	0	include/keepkey/board/usb.h
61	17	include/keepkey/emulator/libkkemu.h
43	1	include/keepkey/firmware/app_confirm.h
9	1	include/keepkey/firmware/app_layout.h
11	1	include/keepkey/firmware/authenticator.h
12	0	include/keepkey/firmware/binance.h
22	0	include/keepkey/firmware/bip85.h
2	0	include/keepkey/firmware/coins.def
3	1	include/keepkey/firmware/coins.h
72	0	include/keepkey/firmware/dice_input.h
11	4	include/keepkey/firmware/eip712.h
227	0	include/keepkey/firmware/eip712_stream.h
4	0	include/keepkey/firmware/eos.h
80	0	include/keepkey/firmware/erc7730_abi.h
90	0	include/keepkey/firmware/erc7730_abi_stream.h
111	0	include/keepkey/firmware/erc7730_capabilities.h
241	0	include/keepkey/firmware/erc7730_catalog.h
20	0	include/keepkey/firmware/erc7730_condition.h
69	0	include/keepkey/firmware/erc7730_field.h
38	0	include/keepkey/firmware/erc7730_format.h
311	0	include/keepkey/firmware/erc7730_program.h
26	0	include/keepkey/firmware/erc7730_tx.h
277	0	include/keepkey/firmware/erc7730_workflow.h
20	2	include/keepkey/firmware/ethereum.h
34	0	include/keepkey/firmware/ethereum_contracts.h
3	0	include/keepkey/firmware/ethereum_contracts/saproxy.h
27	1	include/keepkey/firmware/ethereum_contracts/thorx.h
3	0	include/keepkey/firmware/ethereum_contracts/zxliquidtx.h
14	3	include/keepkey/firmware/ethereum_tokens.h
38	0	include/keepkey/firmware/fsm.h
98	0	include/keepkey/firmware/hive.h
2	0	include/keepkey/firmware/home_sm.h
54	4	include/keepkey/firmware/mayachain.h
36	1	include/keepkey/firmware/osmosis.h
7	0	include/keepkey/firmware/pin_sm.h
9	0	include/keepkey/firmware/recovery_cipher.h
65	4	include/keepkey/firmware/reset.h
28	2	include/keepkey/firmware/ripple.h
242	0	include/keepkey/firmware/signed_metadata.h
21	1	include/keepkey/firmware/signing.h
16	3	include/keepkey/firmware/signtx_tendermint.h
148	0	include/keepkey/firmware/solana.h
45	1	include/keepkey/firmware/storage.h
30	0	include/keepkey/firmware/tendermint.h
50	5	include/keepkey/firmware/thorchain.h

4	1	include/keepkey/firmware/tiny-json.h
0	7	include/keepkey/firmware/tokens.def
10	1	include/keepkey/firmware/transaction.h
56	0	include/keepkey/firmware/tron.h
3	1	include/keepkey/firmware/txin_check.h
431	0	include/keepkey/firmware/zcash.h
19	0	include/keepkey/rand/rng.h
143	0	include/keepkey/rand/rng_health.h
2	0	include/keepkey/transport/interface.h
27	0	include/keepkey/transport/messages-ethereum.options
56	0	include/keepkey/transport/messages-hive.options
0	1	include/keepkey/transport/messages-osmosis.options
2	0	include/keepkey/transport/messages-ripple.options
12	0	include/keepkey/transport/messages-solana.options
1	0	include/keepkey/transport/messages-thorchain.options
54	0	include/keepkey/transport/messages-zcash.options
16	0	include/keepkey/transport/messages.options
222	113	include/pb.h
512	30	lib/board/confirm_sm.c
163	16	lib/board/draw.c
4	1	lib/board/font.c
13	10	lib/board/keepkey_board.c
16	6	lib/board/keepkey_flash.c
152	9	lib/board/layout.c
162	68	lib/board/messages.c
39	9	lib/board/signatures.c
28	2	lib/board/timer.c
3	0	lib/board/udp.c
50	23	lib/board/usb.c
48	16	lib/board/util.c
7	0	lib/emulator/CMakeLists.txt
302	35	lib/emulator/libkkemu.c
45	5	lib/emulator/setup.c
8	2	lib/emulator/udp.c
56	27	lib/firmware/CMakeLists.txt
281	44	lib/firmware/app_confirm.c
117	4	lib/firmware/app_layout.c
240	131	lib/firmware/authenticator.c
141	14	lib/firmware/binance.c
105	0	lib/firmware/bip85.c
19	1	lib/firmware/coins.c
448	0	lib/firmware/dice_input.c
460	255	lib/firmware/eip712.c
1456	0	lib/firmware/eip712_stream.c
21	8	lib/firmware/eos-contracts/eosio.system.c
117	13	lib/firmware/eos.c
472	0	lib/firmware/erc7730_abi.c
463	0	lib/firmware/erc7730_abi_stream.c
243	0	lib/firmware/erc7730_capabilities.c
1652	0	lib/firmware/erc7730_catalog.c
109	0	lib/firmware/erc7730_condition.c
297	0	lib/firmware/erc7730_field.c
233	0	lib/firmware/erc7730_format.c
877	0	lib/firmware/erc7730_program.c
46	0	lib/firmware/erc7730_tx.c
1097	0	lib/firmware/erc7730_workflow.c
659	119	lib/firmware/ethereum.c
51	11	lib/firmware/ethereum_contracts.c
36	24	lib/firmware/ethereum_contracts/makerdao.c
52	13	lib/firmware/ethereum_contracts/saproxy.c
189	47	lib/firmware/ethereum_contracts/thortx.c
113	83	lib/firmware/ethereum_contracts/zxapliquid.c
154	111	lib/firmware/ethereum_contracts/zxliquidtx.c
201	36	lib/firmware/ethereum_contracts/zxswap.c
80	16	lib/firmware/ethereum_contracts/zxtransERC20.c

21	13	lib/firmware/ethereum_tokens.c
416	19	lib/firmware/fsm.c
58	17	lib/firmware/fsm_msg_binance.h
143	0	lib/firmware/fsm_msg_bip85.h
52	9	lib/firmware/fsm_msg_coin.h
143	35	lib/firmware/fsm_msg_common.h
118	23	lib/firmware/fsm_msg_cosmos.h
27	14	lib/firmware/fsm_msg_crypto.h
24	7	lib/firmware/fsm_msg_debug.h
22	3	lib/firmware/fsm_msg_eos.h
2239	141	lib/firmware/fsm_msg_ethereum.h
491	0	lib/firmware/fsm_msg_hive.h
162	30	lib/firmware/fsm_msg_mayachain.h
210	113	lib/firmware/fsm_msg_osmosis.h
81	9	lib/firmware/fsm_msg_ripple.h
814	153	lib/firmware/fsm_msg_solana.h
125	23	lib/firmware/fsm_msg_tendermint.h
136	29	lib/firmware/fsm_msg_thorchain.h
37	55	lib/firmware/fsm_msg_ton.h
111	15	lib/firmware/fsm_msg_tron.h
1699	0	lib/firmware/fsm_msg_zcash.h
439	0	lib/firmware/hive.c
46	8	lib/firmware/home_sm.c
389	97	lib/firmware/mayachain.c
54	0	lib/firmware/messagemap.def
5	2	lib/firmware/nano.c
270	37	lib/firmware/osmosis.c
77	6	lib/firmware/passphrase_sm.c
47	21	lib/firmware/pin_sm.c
213	90	lib/firmware/recovery_cipher.c
482	96	lib/firmware/reset.c
65	10	lib/firmware/ripple.c
1046	0	lib/firmware/signed_metadata.c
549	59	lib/firmware/signing.c
193	65	lib/firmware/signtx_tendermint.c
538	41	lib/firmware/solana.c
306	42	lib/firmware/storage.c
8	0	lib/firmware/storage.h
180	4	lib/firmware/tendermint.c
318	46	lib/firmware/thorchain.c
5	3	lib/firmware/tiny-json.c
5	3	lib/firmware/ton.c
180	18	lib/firmware/transaction.c
364	1	lib/firmware/tron.c
14	7	lib/firmware/txin_check.c
11	2	lib/firmware/u2f.c
1237	0	lib/firmware/zcash.c
2	1	lib/rand/CMakeLists.txt
97	13	lib/rand/rng.c
348	0	lib/rand/rng_health.c
18	0	lib/transport/CMakeLists.txt
2	1	lib/transport/pb_decode.c
1	1	scripts/build/docker/device/debug.sh
7	2	scripts/build/docker/device/release.sh
1	1	scripts/build/docker/emulator/debug.sh
1	0	scripts/cppcheck-version
32	0	scripts/cppcheck.sh
10	2	scripts/emulator/Dockerfile
111	0	scripts/emulator/capture-dice-flow.py
99	0	scripts/emulator/capture-thor-percent.py
12	0	scripts/emulator/docker-compose.bitcoin-only.yml
14	0	scripts/emulator/docker-compose.yml
19	1	scripts/emulator/firmware-unit.sh
87	65	scripts/emulator/python-keepkey-tests.sh
1	1	scripts/emulator/python-keepkey.Dockerfile

443	0	scripts/emulator/test_stack07_regressions.py
36	0	scripts/emulator/test_stack09_integration.py
141	0	scripts/emulator/test_stack10_regressions.py
623	68	scripts/generate-test-report.py
103	0	scripts/preflight.sh
92	0	scripts/preflight_checks.py
248	0	scripts/test_generate_test_report.py
124	0	scripts/test_preflight_checks.py
91	0	scripts/test_preflight_cppcheck.py
85	0	scripts/tests/test_usb_receive_callbacks.py
36	0	scripts/verify-token-def.py
1	0	tools/CMakeLists.txt
124	0	tools/check_sram_budget.py
33	11	tools/emulator/CMakeLists.txt
20	0	tools/erc7730-validate/CMakeLists.txt
76	0	tools/erc7730-validate/main.c
3	1	tools/firmware/CMakeLists.txt
10	0	tools/firmware/keepkey.ld
14	9	tools/firmware/keepkey_main.c
10	0	tools/sram-budgets.json
251	0	tools/verify_dice_seed.py
1	1	unittests/board/CMakeLists.txt
490	2	unittests/board/board.cpp
17	2	unittests/crypto/CMakeLists.txt
500	0	unittests/crypto/bip340.cpp
21	0	unittests/crypto/emulator_random.c
55	7	unittests/firmware/CMakeLists.txt
48	0	unittests/firmware/app_confirm.cpp
302	0	unittests/firmware/authenticator.cpp
151	0	unittests/firmware/binance.cpp
28	2	unittests/firmware/coins.cpp
141	0	unittests/firmware/confirm_test_utils.cpp
166	8	unittests/firmware/cosmos.cpp
116	0	unittests/firmware/dice.cpp
194	0	unittests/firmware/eip712.cpp
1076	0	unittests/firmware/eip712_stream.cpp
90	3	unittests/firmware/eos.cpp
50	0	unittests/firmware/eos_authorization.cpp
16	0	unittests/firmware/eos_authorization_probe.c
174	0	unittests/firmware/erc7730_abi.cpp
371	0	unittests/firmware/erc7730_abi_stream.cpp
1901	0	unittests/firmware/erc7730_catalog.cpp
105	0	unittests/firmware/erc7730_condition.cpp
304	0	unittests/firmware/erc7730_field.cpp
224	0	unittests/firmware/erc7730_format.cpp
411	0	unittests/firmware/erc7730_program.cpp
74	0	unittests/firmware/erc7730_tx.cpp
207	0	unittests/firmware/erc7730_workflow.cpp
709	1	unittests/firmware/ethereum.cpp
1204	0	unittests/firmware/fsm.cpp
783	0	unittests/firmware/hive.cpp
20	0	unittests/firmware/kkconfirm_driver.h
54	0	unittests/firmware/liquidity_key_probe.c
390	7	unittests/firmware/mayachain.cpp
11	0	unittests/firmware/messages_probe.c
35	0	unittests/firmware/nanopb_bounds.cpp
153	0	unittests/firmware/osmosis.cpp
46	0	unittests/firmware/recovery.cpp
37	0	unittests/firmware/ripple.cpp
297	0	unittests/firmware/rng_health.cpp
193	0	unittests/firmware/setup_ceremony.cpp
1804	0	unittests/firmware/signed_metadata.cpp
143	0	unittests/firmware/signing.cpp
969	15	unittests/firmware/solana.cpp
316	29	unittests/firmware/storage.cpp

129	0	unittests/firmware/storage_cipher_probe.c
24	0	unittests/firmware/storage_cipher_probe.h
140	0	unittests/firmware/storage_passphrase.cpp
28	0	unittests/firmware/test_board.cpp
350	21	unittests/firmware/thorchain.cpp
173	0	unittests/firmware/transaction.cpp
604	0	unittests/firmware/tron.cpp
169	0	unittests/firmware/usb_rx.cpp
2204	0	unittests/firmware/zcash.cpp